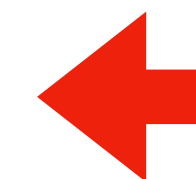
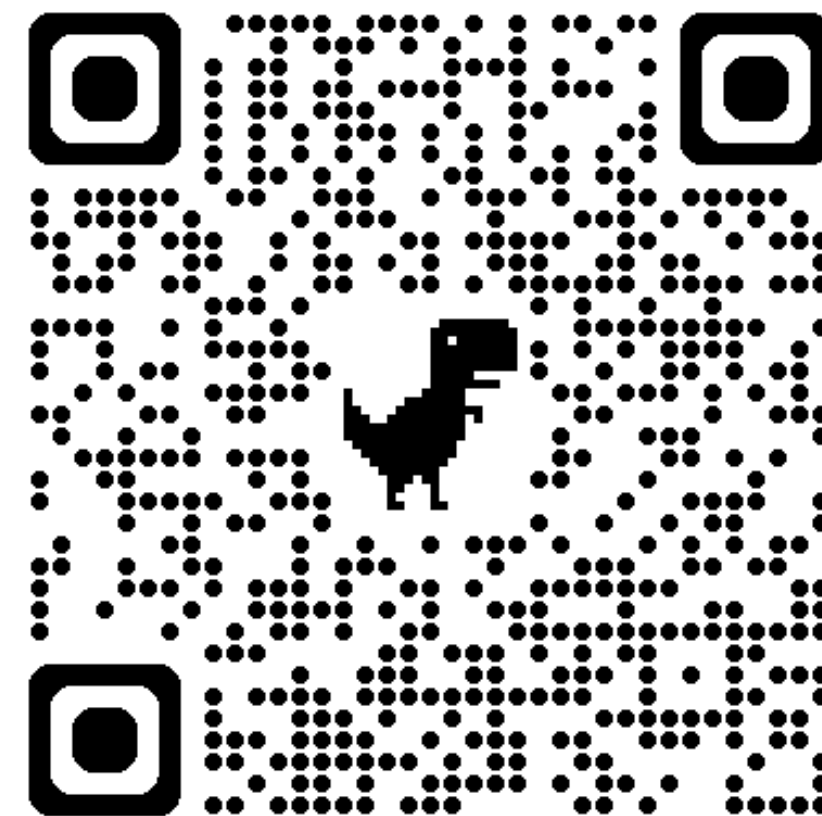


Academic Paper Writing in Artificial Intelligence

Evaluation Metrics and Baselines

Instructor: Ying-Jia Lin
September 24th, 2026



Couse Website
Download slide here!

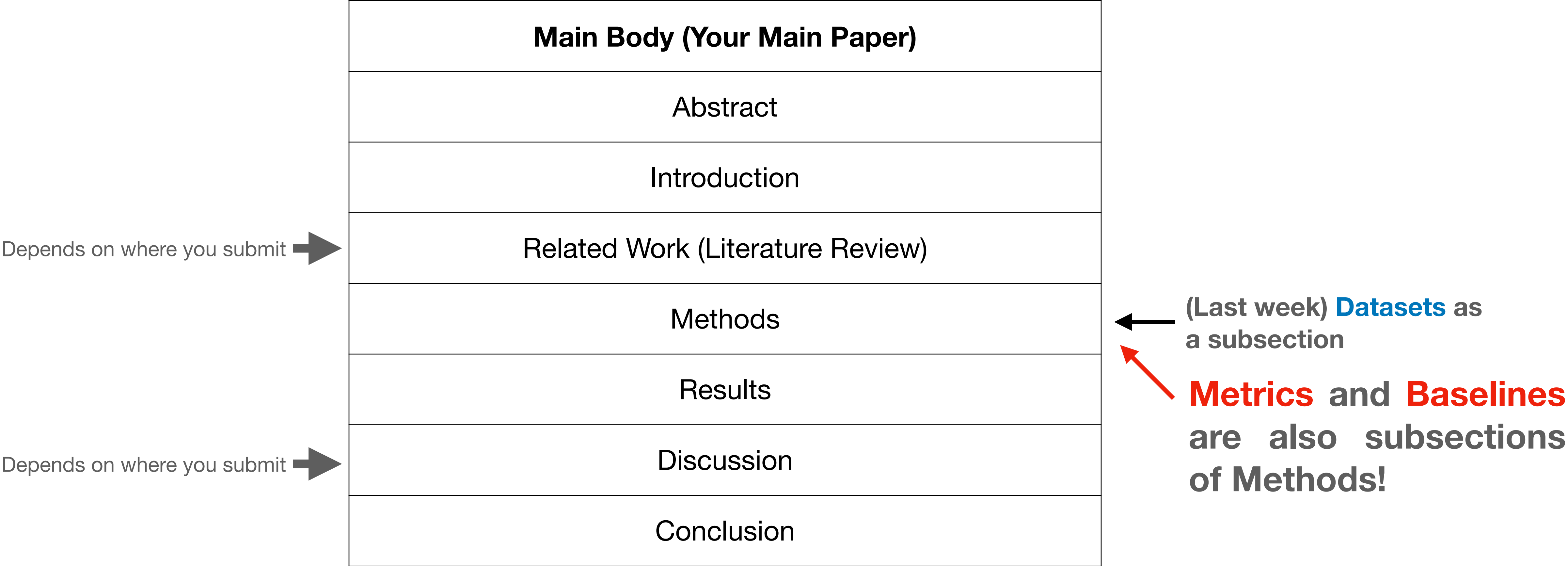
Outline

- Introduction to “Evaluation Metrics” (10 min)
- How to write the "Evaluation Metrics" subsection (20 min)
- **Your turn to revise and write (45 min, including break)**
- Introduction to “Baselines” (10 min)
- How to write the "Baselines" subsection (20 min)
- LaTeX: `\citet` & `\citep`, metric macros (5 min)
- **Your turn to revise and write (40 min)**

Where we are

	Main Body (Your Main Paper)	
	Abstract	
	Introduction	
Depends on where you submit →	Related Work (Literature Review)	
	Methods	← (Last week) Datasets as a subsection
	Results	
Depends on where you submit →	Discussion	
	Conclusion	

Where we are



W2 Assignment Summary (1)

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- The dataset statistics are missing.

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W2 Assignment Summary (1)

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 - Some immediate numbers are missing.
 - The removal definition is not clear.
- Avoid the ambiguous terms like “approximate” or “about”.

W2 Assignment Summary (2) & W3 Introduction

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- Some students' assignments contain my highlight - "Week03".

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Evaluation Metrics and Baselines

You have good results after experiments ...

- Before you start writing paper, you claim your model reaches **92.4**.

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Question: Why do the readers believe your paper?

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Question: Why do the readers believe your paper?

➡ “Metrics” tells readers “How the numbers were computed”.

You have good results after experiments ...

- Before you start writing paper, you claim your model reaches **92.4**.

Question: Why do the readers believe your paper?

➡ “Metrics” tells readers “How the numbers were computed”.

➡ “Baselines” tells readers “What the numbers should be compared with” - so the reader can see the advantage of your method.

Reviewers are your most important readers

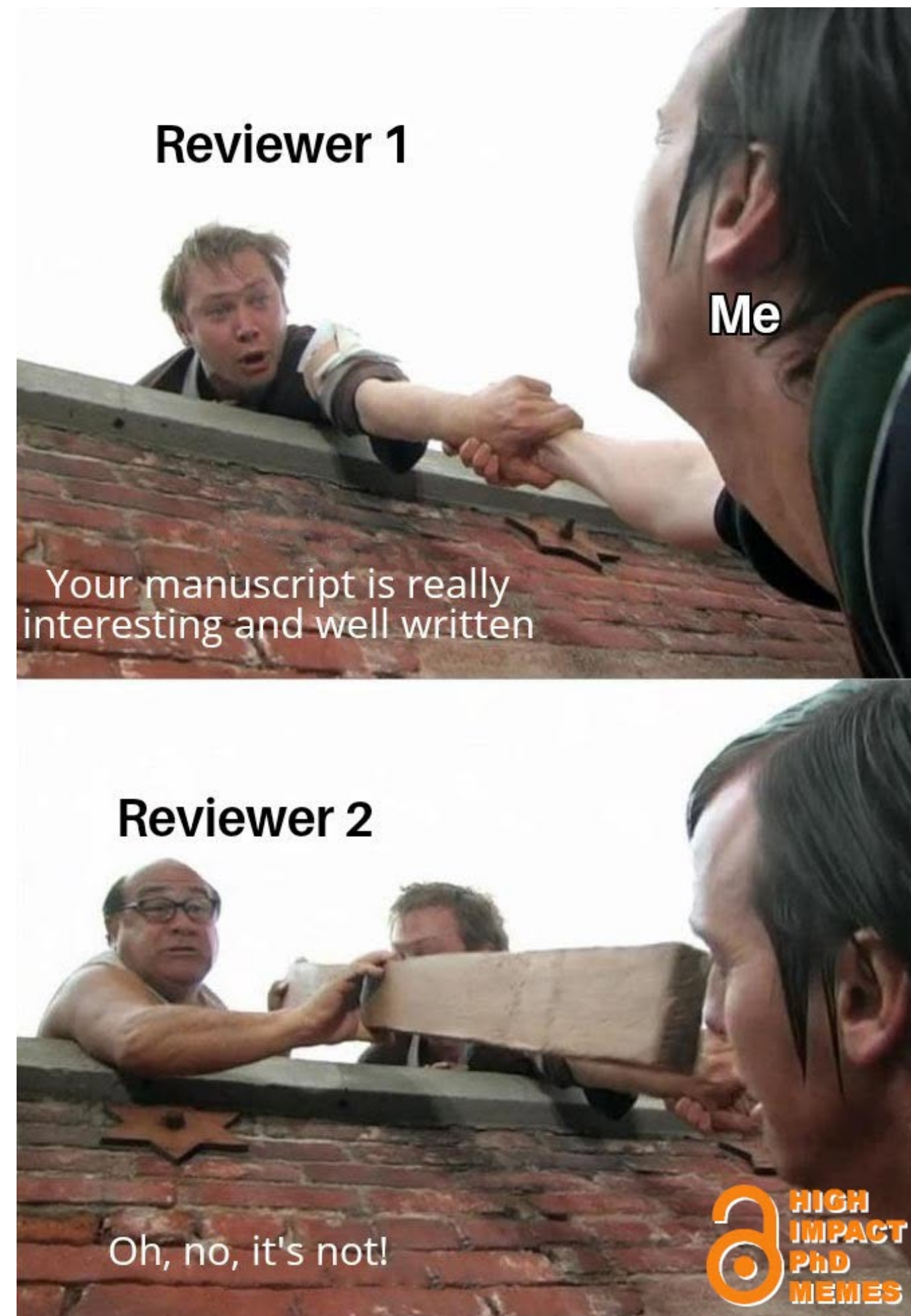


Figure source: <https://www.facebook.com/MemingPhD/posts/3122144371363205/>

Evaluation Metrics

Core Idea

- A metric is **how you turn your claim into a number.**

(Example)

- Your claim: "Our model produces more faithful clinical summaries."

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⬆ Your readers: what number changes when faithfulness improves?

Core Idea

- A metric is **how you turn your claim into a number.**

(Example)

- Your claim: "Our model produces more faithful clinical summaries."

↑ Your readers: what number changes when faithfulness improves?

↑ Why do we need to measure faithfulness?

Example: Language Generation

	ROUGE-1	ROUGE-2	ROUGE-L	BLEURT	BERTScore	BARTScore
LLMs Results						
GPT-4o	41.2	15.9	26.0	54.7	69.4	-2.78
Gemini-1.5-pro	41.5	16.7	27.2	53.9	69.7	-2.67
Mistral-Large	45.1	18.8	29.4	55.9	71.2	-2.48
LLMs Average	42.0	17.3	27.2	54.5	69.7	-2.59
Dataset: LLMs-9						
UnRank	43.8 $\pm$ 0.83	20.5 $\pm$ 0.46	30.1 $\pm$ 0.93	53.3 $\pm$ 0.27	<u>69.9</u> $\pm$ 0.27	-2.59 $\pm$ 0.47
ROUGE	43.2 $\pm$ 0.23	<u>20.4</u> $\pm$ 0.57	29.9 $\pm$ 0.22	51.4 $\pm$ 0.40	69.0 $\pm$ 0.05	-2.58 $\pm$ 0.02
BERTScore	43.9 $\pm$ 0.59	20.0 $\pm$ 0.58	30.1 $\pm$ 0.81	<u>52.2</u> $\pm$ 0.50	69.5 $\pm$ 0.32	-2.46 $\pm$ 0.06
BLANC	<u>44.1</u> $\pm$ 0.54	20.3 $\pm$ 0.47	<u>30.2</u> $\pm$ 0.35	50.1 $\pm$ 0.64	69.4 $\pm$ 0.52	-2.34 $\pm$ 0.13
GPTRank	43.8 $\pm$ 1.12	20.0 $\pm$ 1.66	30.1 $\pm$ 1.17	51.4 $\pm$ 1.36	69.7 $\pm$ 0.66	<u>-2.36</u> $\pm$ 0.05
SCURank	44.8 $\pm$ 0.28	20.5 $\pm$ 0.81	30.6 $\pm$ 0.86	51.7 $\pm$ 0.78	70.0 $\pm$ 0.27	-2.34 $\pm$ 0.06

Example: Language Generation

Why so many metrics?

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Example: Language Generation

Metrics for overlaps

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Example: Language Generation

Metrics for overlaps

Metrics for **semantics**

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Three steps to write a Metrics subsection

1. Announce — name the metric set, name the primary metric, and say why.

2. Specify — pin down the exact variant and implementation.
(when it matters)

3. Report — seeds, mean $\pm$ std, confidence intervals, significance test.

Step 1 — Announce, and pick a primary metric

We evaluate all models with macro-averaged F1, AUROC, and AUPRC. **We use AUPRC as the primary metric because positive cases constitute only 3.1% of the cohort.**

Why naming a primary metric matters:

- It stops you from cherry-picking whichever metric happens to win.
- It tells the reviewer **which column** of Table 2 carries your argument.
- It forces you to decide what your paper is actually about.

Step 2 — Specify the variant (when it matters)

- A metric name alone is not reproducible information:

You wrote	The reviewer needs
F1	micro / macro / weighted? span-level or token-level?
BLEU	SacreBLEU signature; tokenizer; case-sensitivity
AUC	AUROC or AUPRC?
Sensitivity	at which operating threshold? chosen on which split?
...	...

Step 2 in the concrete way

See, Abigail, Peter J. Liu, and Christopher D. Manning. "Get to the point: Summarization with pointer-generator networks." ACL 2017.

We obtain our ROUGE scores using the pyrouge package.⁴ We also evaluate with the METEOR metric (Denkowski and Lavie, 2014), both in exact match mode (rewarding only exact matches between words) and full mode (which additionally rewards matching stems, synonyms and paraphrases).⁵

⁴pypi.python.org/pypi/pyrouge/0.1.3

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Footnote (may include toolkit versions!)

Three steps to write a Metrics subsection

1. Announce — name the metric set, name the primary metric, and say why.

2. Specify — mention the exact variant and implementation (details).

Now we are here.

3. Report — seeds, mean $\pm$ std, confidence intervals, significance test.

Three steps to write a Metrics subsection

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↓ **Now we are here.**

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Step 3 — Report protocols (Example 1)

Ghosh, Sayontan, et al. "PASTA: A dataset for modeling PArticipant STates in narratives." Transactions of the Association for Computational Linguistics 11 (2023): 1283-1300.

- An example without statistical tests

We evaluated model performance with standard accuracy and contrastive accuracy. In contrastive accuracy, the model gets a point only if it makes correct predictions for both inferred and the counterfactual states for a story. **For all models, we train with five random seeds and report their average performance with standard deviation.**

Step 3 — Report protocols (Example 2)

- A concise example

Zhou, Nan, et al. "STeP-Diff: Spatio-Temporal Physics-Informed Diffusion Models for Mobile Fine-Grained Pollution Forecasting." IEEE Transactions on Knowledge and Data Engineering (2026).

We conducted statistical significance testing on 12 evaluation metrics by comparing our model with the second-best baseline. For each metric, we performed 10 independent runs and applied a paired t-test for analysis.

Step 3 — Report protocols (Example 3)

Li, Jie, et al. "Bridging the Generalist–Subspecialist Gap with GPT-5-Thinking: Dual-Center Evaluation in Orbital and Head-and-Neck Tumor MRI Reports." *Radiology* 320.3 (2026): e253579.

Continuous variables were summarized as mean $\pm$ SD for normally distributed data and as median (IQR, range) for skewed data; categorical variables were summarized as counts and percentages. For group comparisons, continuous variables were analyzed using independent-samples t tests or Mann-Whitney U tests. For categorical variables, proportions were compared using χ^2 tests for independent groups and McNemar tests for paired data.

Sentence Bank

Type	Example
Announcement	"Following \citet{...}, we evaluate with X and Y."
Reason	"We adopt X as the primary metric because <property of the task>."
Protocol	"Unless otherwise stated, all numbers are averaged over five seeds (mean $\pm$ std)."
Specification	"Metrics are computed with <toolkit> v*<version>*. \footnote{\url{...}}"
Announcement (more details)	"Higher is better for all metrics except X, for which lower is better."
Protocol	"Statistical significance is assessed with a paired bootstrap test ($p < 0.05$)."
Announcement	"We additionally conduct a human evaluation on 200 randomly sampled outputs (§5.3)."

If you use human evaluation or an LLM judge

- Human evaluation — state:
 - number of annotators, their expertise, number of items, inter-annotator agreement (if needed), descriptions for a specific metric (if needed), and so on.
- LLM-as-a-judge — state:
 - exact model version (and date), the full prompt (appendix), how you controlled position/verbosity bias (if needed), and so on.

Example for Human Evaluation

Wang, Bo-Jyun, Ying-Jia Lin, and Hung-Yu Kao. "SCURank: Ranking Multiple Candidate Summaries with Summary Content Units for Enhanced Summarization." Findings of ACL 2026.

Number of
items

Number of
humans

Specific
metrics

We randomly sampled 30 articles from the test sets. For each article, we presented the source text and pairs of summaries generated by the two models to three independent annotators recruited via Amazon Mechanical Turk (MTurk). Annotators were asked to assess the summaries in two ways: (1) scoring them on a Likert scale (1-5) across three dimensions: Faithfulness, Conciseness, and Completeness; and (2) providing an overall preference (Win/Tie/Loss). The details of the quality of annotators and evaluation criteria are provided in the Appendix G.

Your Turn

Baselines

Core Idea

- A baseline is **an argument**, not just a list of items.
- Your baseline set must argue three things at once:
 1. The task is non-trivial.
 2. You advance the state of the art.
 3. The gain comes from your idea — not from more data, more parameters, or more tuning.

Every baseline plays one of the three roles

Role	Purpose	Example
Easy Baseline	Shows the task is not too easy	majority class, random, rule-based, retrieval-only
SOTA	You advance the field	the strongest published method on this dataset
Control	Isolate your contribution	same backbone without your module; same compute budget

An easy baseline for text summarization

https://edition.cnn.com/2007/SHOWBIZ/Movies/07/23/potter.radcliffe.reut/index.html?iref=nextin

Life's magic as Daniel Radcliffe turns 18

- Story Highlights
- Harry Potter star Daniel Radcliffe gets £20M fortune as he turns 18 Monday
- Young actor says he has no plans to fritter his cash away
- Radcliffe's earnings from first five Potter films have been held in trust fund
- [Next Article in Entertainment »](#)

□

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- ☐ [VIDEO](#)
- ☐ [PHOTOS](#)
- ☐ [INTERACTIVE](#)

☐

Decrease font Decrease font
Enlarge font Enlarge font

LONDON, England (Reuters) -- Harry Potter star Daniel Radcliffe gains access to a reported £20 million (\$41.1 million) fortune as he turns 18 on Monday, but he insists the money won't cast a spell on him.

art.harry.potter.jpg

Daniel Radcliffe as Harry Potter in "Harry Potter and the Order of the Phoenix"

[more photos »](#)

To the disappointment of gossip columnists around the world, the young actor says he has no plans to fritter his cash away on fast cars, drink and celebrity parties.

"I don't plan to be one of those people who, as soon as they turn 18, suddenly buy themselves a massive sports car collection or something similar," he told an Australian interviewer earlier this month. "I don't think I'll be particularly extravagant.

"The things I like buying are things that cost about 10 pounds -- books and CDs and DVDs."

LEAD-3

The 3 steps of a Baselines subsection

1. Describe — group them into families, then one sentence per baseline: what it is + citation + why it is here.
 2. Fairness — identical splits, backbone, and tuning budget; which numbers you re-ran and which you copied.
 3. Exclusions — which obvious baselines you left out, and why.
- Target length: 200–300 words.

Step 1 - Group them. Not write a flat list.

✗ We compare our method with SVM, BERT, RoBERTa, T5, GPT-3.5, Method A, Method B, and Method C.

✓ We compare our method against three families of baselines:

- (i) classical models (TF-IDF + SVM), which establish task difficulty;
- (ii) fine-tuned encoders (BERT-base, RoBERTa-large), the dominant paradigm for this task; and (iii) prompted LLMs (GPT-4o, Llama-3-70B), which represent the current strongest published results.

Step 2 - The fairness statement (write it explicitly!)

- Do not make the reviewer guess. State it:

✓ All baselines use the same training/validation/test splits (see the Datasets subsection) and the same RoBERTa-large backbone. For each baseline we tune the learning rate and batch size over the same grid with an identical budget of 20 trials. Numbers marked † are copied from the original papers; all others are obtained by re-running the officially released code in our environment.

Step 3 - Answer the reviewer before they ask

- Optional but important

✓ We do not compare against \citet{...}, as their method requires patient-level genomic features that are unavailable in our cohort. We also exclude closed-source clinical LLMs, since our institutional data-use agreement prohibits transmitting records to external APIs.

- This turns a possible "missing baseline" rejection into a sign that you **thought about it carefully**.

Sentence bank — Baselines

- "We compare \method against three families of baselines."
- "\textbf{X} \citep{...} is a <one-clause description>; we include it because <role>."
- "All models share the same backbone, splits, and hyperparameter search budget."
- "Results marked † are taken from the original publications; all other results are reproduced by us."
- "For a controlled comparison, we additionally report a variant of our model without <component>."
- "We do not compare with X because <concrete, technical reason>."
- "To our knowledge, \citet{...} is the strongest previously published result on this benchmark."

LaTeX: `\citet` vs. `\citep`

- Use ren2026 as an example:

Command	Description	Example
<code>\citet{ren2026}</code>	Textual citation	Ren et al. (2026)
<code>\citep{ren2026}</code>	Parenthetical citation	(Ren et al., 2026)
<code>\citet*{ren2026}</code>	Same as <code>\citet</code> , but if there are several authors, all names will be printed	Ren, Zhang, Zhang, Lang, Zong, and Shen (2026)
<code>\citep*{ren2026}</code>	Same as <code>\citep</code> , but if there are several authors, all names will be printed	(Ren, Zhang, Zhang, Lang, Zong, and Shen, 2026)
<code>\citeauthor{ren2026}</code>	Prints only the author name(s)	Ren et al.
<code>\citeyear{ren2026}</code>	Prints only the year	2026

Your Turn

NEXT WEEK ATTENTION!!

- Next week we will making a method overview figure and write a caption.
- Please prepare your method sketch (plain text or text in items).

Thank you for your participation 